Customer Retention Intelligence System

Telecom Customer Churn Prediction & Analysis | Data Analyst Internship Project

Ashritha Anandal | ashritha.a05@gmail.com

1. Introduction

Customer churn - the rate at which subscribers cancel their service is one of the most critical business metrics in the telecom industry. With acquisition costs typically five times higher than retention costs, even a 1% reduction in monthly churn can translate to millions in saved revenue. This project addresses that challenge by building an end-to-end **Customer Retention Intelligence System**: a complete analytical pipeline that moves from raw transactional data through exploratory analysis, predictive modelling, and explainability - culminating in a live web application that any business user can operate to assess customer risk in real time.

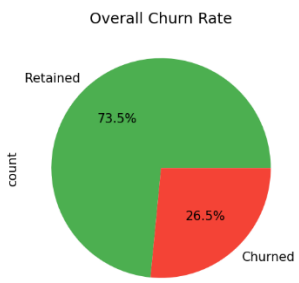
2. Abstract

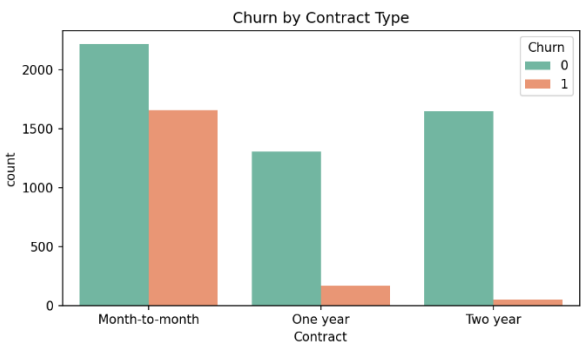
Using the IBM Telco Customer Churn dataset (7,043 customers, 21 features) sourced from Kaggle, this project delivers a full data analysis and machine learning pipeline. An XGBoost classification model was trained to predict whether a customer will churn, achieving a **ROC-AUC score of ~0.84** on a held-out test set. SHAP were used to move beyond accuracy providing per-customer explanations of *why* the model predicts churn. Customers were further segmented into three actionable risk tiers: **At Risk**, **Watch**, and **Loyal**. All findings are surfaced through a deployed Streamlit web application where a user can enter any customer profile and receive an instant churn probability score alongside personalised retention recommendations.

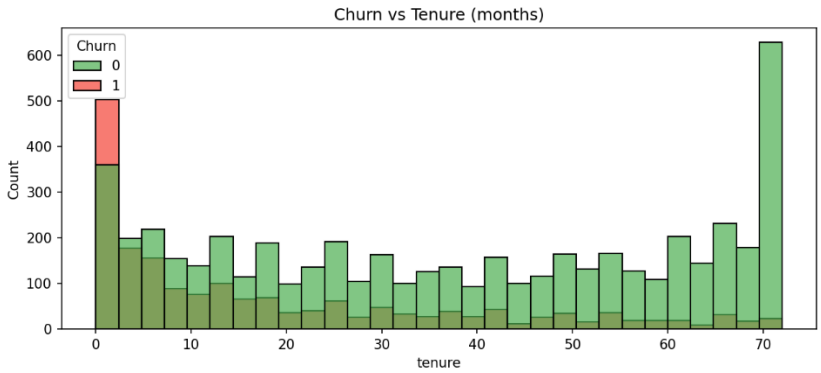
3. Tools Used

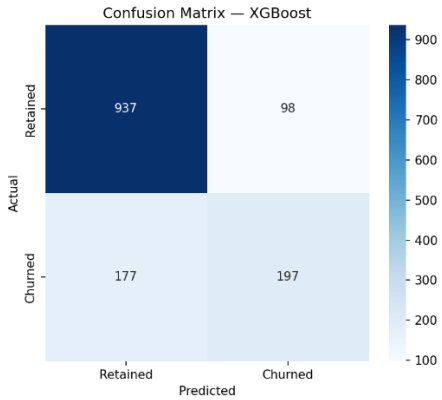
Tool / Library	Purpose
Python (Pandas, NumPy)	Data cleaning, feature engineering, encoding
Matplotlib & Seaborn	EDA charts, distribution plots, heatmaps
Scikit-learn	Train/test split, evaluation metrics
XGBoost	Primary classification model
SHAP	Model explainability - feature impact analysis
Google Colab	Cloud notebook environment for all code
Streamlit (streamlit.io)	Live web app - real-time churn prediction UI

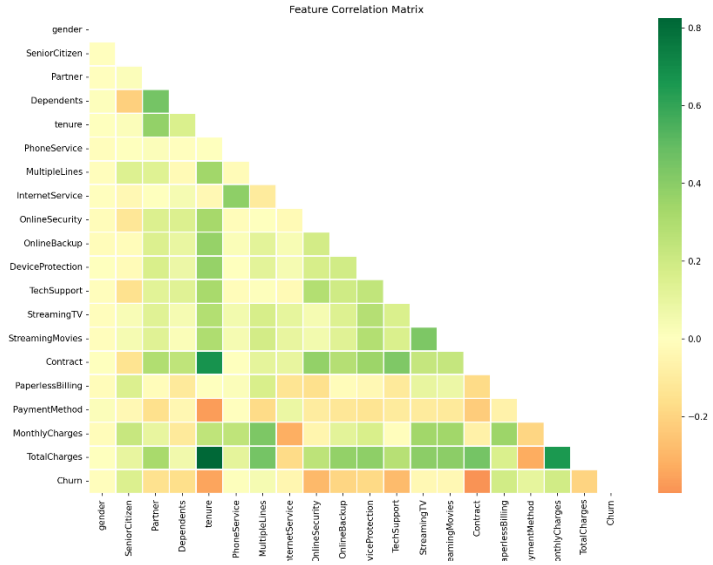
4. Key Visualisations

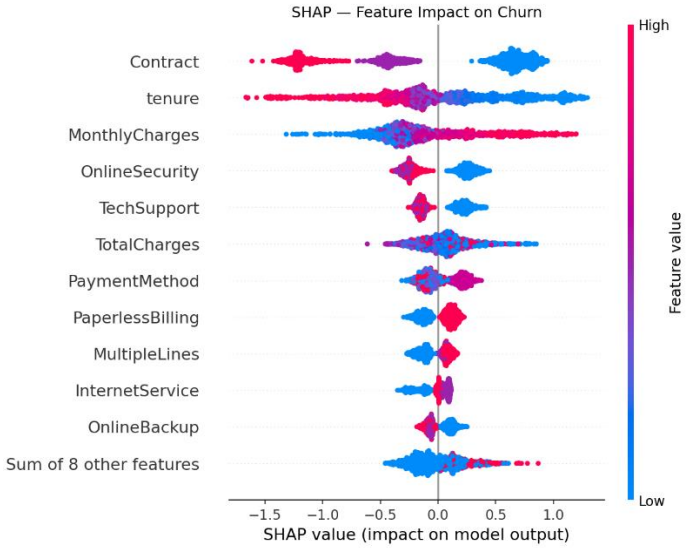


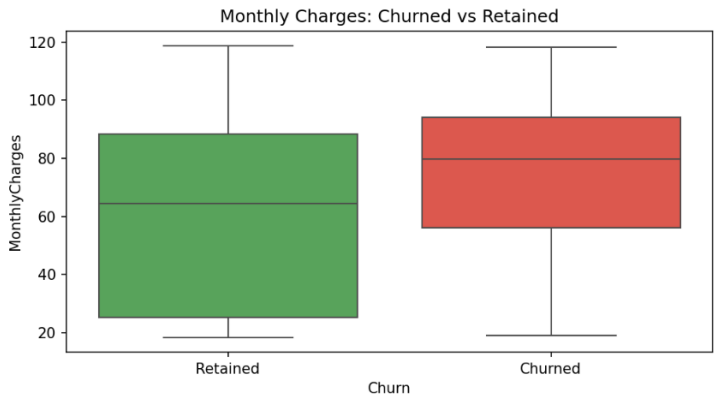












5. Steps Involved in Building the Project

1	**Data Acquisition**	Downloaded the IBM Telco Customer Churn dataset from Kaggle (7,043 rows, 21 features).
2	**Data Cleaning**	Converted TotalCharges to numeric, imputed 11 missing values with the median, dropped the customerID column, and binary-encoded the Churn target.
3	**Exploratory Data Analysis (EDA)**	Generated 7 visualisations covering churn distribution, contract type breakdown, tenure histogram, monthly charges boxplot, and a full correlation heatmap.
4	**Feature Engineering**	Label-encoded all 16 categorical columns, derived TotalCharges from tenure x MonthlyCharges, and confirmed zero remaining nulls before modelling.

5	**Model Training**	Trained an XGBoost classifier (200 estimators, lr=0.05, max_depth=4) on an 80/20 stratified split. Achieved ROC-AUC of ~0.84.
6	**Model Evaluation**	Evaluated using classification report, confusion matrix, and ROC-AUC score. SHAP beeswarm plot used to explain individual predictions.
7	**Customer Segmentation**	Assigned each test-set customer to a risk tier — At Risk (>=70%), Watch (40-70%), Loyal (<40%) — and visualised segments on a tenure vs charges scatter plot.
8	**Model Export**	Serialised the trained XGBoost model and feature column list as .pkl files using pickle, then downloaded from Colab.
9	**Web App Deployment**	Built a Streamlit app (app.py) that loads the .pkl model, accepts customer inputs via sidebar sliders/dropdowns, and returns live churn probability + personalised retention tips. Deployed free on streamlit.io.

6. Conclusion

This project demonstrates a complete, production-oriented data analytics pipeline. The analysis confirmed that contract type, tenure, and monthly charges are the strongest predictors of churn — month-to-month customers with high bills and short tenure are disproportionately at risk. The SHAP analysis added a critical layer of business by making model decisions interpretable to non-technical stakeholders. The three-tier customer segmentation (At Risk / Watch / Loyal) translates model output into actionable retention strategies.

Finally, the live Streamlit web app bridges the gap between data science and business operations — allowing any team member to assess a customer's churn risk in real time without any coding knowledge. Together, these components represent the kind of end-to-end thinking that differentiates a data analyst from a report generator.

Web app link: https://churn-predictor-ashrithaxx.streamlit.app/